

Q3 Applying conservation of momentum,

$$0 = m_1 v_1 + (-m_2 v_2)$$

$$m_1 v_1 = m_2 v_2$$

$$\frac{v_1}{v_2} = \frac{m_2}{m_1}$$

Answer: C

Q4 By resolving each forces into its vertical and horizontal components, only option B is most